

Spanning Connectivity and Spanning Laceability of Hypercube-like Networks

Cheng-Kuan Lin and Jimmy J. M. Tan

Department of Computer Science,
National Chiao Tung University, Hsinchu, Taiwan 30010, R.O.C.
{cklin, jmtan}@cis.nctu.edu.tw

D. Frank Hsu

Department of Computer and Information Science,
Fordham University, New York, NY 10023, USA
hsu@trill.cis.fordham.edu

Lih-Hsing Hsu

Department of Computer Science and Information Engineering,
Providence University, Taichung, Taiwan 43301, R.O.C.
lhhsu@pu.edu.tw

Abstract

Let u and v be any two distinct nodes of an undirected graph G , which is k -connected. For $1 \leq w \leq k$, a w -container $C(u, v)$ of a k -connected graph G is a set of w -disjoint paths joining u and v . A w -container $C(u, v)$ of G is a w^* -container if it contains all the nodes of G . A graph G is w^* -connected if there exists a w^* -container between any two distinct nodes. A bipartite graph G is w^* -laceable if there exists a w^* -container between any two nodes from different parts of G . In this study, we show that every graph in n -dimensional bipartite graph B'_n is w^* -laceable for every w , $1 \leq w \leq n$. It is shown that a constructed an n -dimensional non-bipartite graph H can not be 4^* -connected. In addition, we show that every graph in n -dimensional non-bipartite graph is w^* -connected for every w , $1 \leq w \leq 3$.

1 Introduction

1.1 Definitions

For graph definitions and notation we follow [4]. $G = (V, E)$ is a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that V is the *node set* and E is the *edge*

set. We use $n(G)$ to denote $|V|$. Two nodes u and v are *adjacent* if (u, v) is an edge of G . For a node u , $N_G(u)$ denotes the *neighborhood* of u which is the set $\{v \mid (u, v) \in E\}$. For any node u of V , we denote the *degree* of u by $\deg_G(u) = |N_G(u)|$. A graph G is k -regular if $\deg_G(u) = k$ for every node u in G . A *path* P between nodes v_1 and v_k is a sequence of adjacent nodes, $\langle v_1, v_2, \dots, v_k \rangle$, in which the nodes $v_1, v_2, \dots, v_k$ are distinct except that possibly $v_1 = v_k$. We use P^{-1} to denote the path $\langle v_k, v_{k-1}, \dots, v_1 \rangle$. The *length* of P , $l(P)$, is the number of edges in P . We also write the path P as $\langle v_1, v_2, \dots, v_i, Q, v_j, v_{j+1}, \dots, v_k \rangle$, where Q is the path $\langle v_i, v_{i+1}, \dots, v_j \rangle$. Hence, it is possible to write a path as $\langle v_1, v_2, Q, v_2, v_3, \dots, v_k \rangle$ if $l(Q) = 0$. Let $I(P) = V(P) - \{v_1, v_k\}$ be the set of the internal nodes of P . A set of paths $\{P_1, P_2, \dots, P_k\}$ are *internally node-disjoint* (abbreviated as *disjoint*) if $I(P_i) \cap I(P_j) = \emptyset$ for any $i \neq j$. A path is a *hamiltonian path* if it contains all nodes of G . A graph G is *hamiltonian connected* if there exists a hamiltonian path joining any two distinct nodes of G [16]. A *cycle* is a path with at least three nodes such that the first node is the same as the last one. A *hamiltonian cycle* of G is a cycle that traverses every node of G . A graph is *hamiltonian* if it has a hamiltonian cycle. A graph G is *bipartite* if its node set can be partitioned into two subsets V_1 and V_2 such that

every edge connects nodes between V_1 and V_2 . A bipartite graph G is *hamiltonian laceable* if there is a hamiltonian path of G joining any two nodes from distinct bipartition [18]. A bipartite graph G is *k-edge fault hamiltonian laceable* if $G - F$ is hamiltonian laceable for any edge subset F of G with $|F| \leq k$.

A graph G is *k-connected* if there exists a set of k internally disjoint paths $\{P_1, P_2, \dots, P_k\}$ between any two distinct nodes u and v . A subset S of $V(G)$ is a *cut set* if $G - S$ is disconnected. A *w-container* of G between two distinct nodes u and v is a set of w internally disjoint paths between u and v . The concepts of a container and of a wide distance were proposed by Hsu [12] to evaluate the performance of communication for an interconnection network. The *connectivity* of G , $\kappa(G)$, is the minimum number of nodes whose removal leaves the remaining graph disconnected or trivial. Hence, a graph G is *k-connected* if $\kappa(G) \geq k$. It follows from Menger's Theorem [15] that there is a *w-container* for $w \leq k$ between any two distinct nodes of G if G is *k-connected*.

1.2 w^* -connected graphs and w^* -laceable graphs

In this paper, we are interested in a specific type of container. We say that a *w-container* $C(u, v)$ is a *w^* -container* if every node of G is on some path in $C(u, v)$. A graph G is said to be *w^* -connected* if there exists a *w^* -container* between any two distinct nodes u and v . Obviously, we have the following remarks:

Remark 1 (1.a) a graph G is *1*-connected* if and only if it is *hamiltonian connected* [16], (1.b) a graph G is *2*-connected* if it is *hamiltonian*, and (1.c) an *1*-connected* graph except K_1 and K_2 is *2*-connected*.

Using our definition of a *w^* -connected* graph, the globally *3*-connected* graphs proposed by Albert et al. [3] are 3-regular *3*-connected* graphs. Assume that the graph G is *w^* -connected* where $w \leq \kappa(G)$. The *spanning connectivity* of a graph G , $\kappa^*(G)$, is the largest integer k such that G is *i^* -connected* for every i , $1 \leq i \leq k$. A graph G is *super spanning connected* if $\kappa^*(G) = \kappa(G)$. In such case, the number $\kappa^*(G) = \kappa(G)$ is called the *super spanning connectivity* of G . In [14, 20], some families of graphs are proved to be *super spanning connected*.

Let G be a bipartite graph with bipartition V_1 and V_2 such that $|V_1| \geq |V_2|$. Suppose that there

exists a *w^* -container* $C(u, v) = \{P_1, P_2, \dots, P_w\}$ in G joining u to v with $u, v \in V_1$. Obviously, the number of nodes in P_i is $2t_i + 1$ for some integer t_i . There are $t_i - 1$ nodes of P_i in V_1 other than u and v , and t_i nodes of P_i in V_2 . As a consequence, $|V_1| = \sum_{i=1}^w (t_i - 1) + 2$ and $|V_2| = \sum_{i=1}^w t_i$. Therefore, any bipartite graph G with $\kappa(G) \geq 3$ is not *w^* -connected* for any w , $3 \leq w \leq \kappa(G)$.

For this reason, a bipartite graph is said to be *w^* -laceable* if there exists a *w^* -container* between any two nodes from different partite sets for some w , $1 \leq w \leq \kappa(G)$. Obviously, any bipartite *w^* -laceable* graph with $w \geq 2$ has the equal size of bipartition. We have the following remarks:

Remark 2 (2.a) an *1*-laceable* graph is also known as *hamiltonian laceable* graph [18], (2.b) a graph G is *2*-laceable* if and only if it is *hamiltonian*, and (2.c) an *1*-laceable* graph except K_1 and K_2 are *2*-laceable*.

The *spanning laceability* of a bipartite graph G , $\kappa^{*L}(G)$, is the largest integer k such that G is *i^* -laceable* for every i , $1 \leq i \leq k$. A graph G is *super spanning laceable* if the number $\kappa^{*L}(G) = \kappa(G)$. Recently, Chang et al. [5] proved that the n -dimensional hypercube Q_n is *super spanning laceable* for every positive integer n . It was proved in [14] that the n -dimensional star graph S_n is *super spanning laceable* if and only if $n \neq 3$.

1.3 Hypercube-like graphs H'_n

Graph containers do exist in engineering design information and telecommunication networks or in biological and neural systems ([2, 12] and its references). The study of *w-container*, *w-wide distance*, and their *w^* -versions* play a pivotal role in the design and the implementation of parallel routing and efficient information transmission in large scale networking systems. In bioinformatics and neuroinformatics, the existence as well as the structure of a *w^* -container* signifies the cascade effect in the signal transduction system and the reaction in a metabolic pathway.

Among all interconnection networks proposed in the literature, the hypercube Q_n is one of the most popular topologies [5, 13]. However, the hypercube does not have the smallest diameter for its resources. Various networks are proposed by twisting some pairs of links in hypercubes [1, 8, 10, 11]. Because of the lack of the unified perspective on these variants, results of one topology are hard to be extended to others. To make

a unified study of these variants, Vaidya et al. introduced the class of hypercube-like graphs [21]. We denote these graphs as H' -graphs. The class of H' -graphs, consisting of simple, connected, and undirected graphs, contains most of the hypercube variants.

Let $G_0 = (V_0, E_0)$ and $G_1 = (V_1, E_1)$ be two disjoint graphs with the same number of nodes. A 1-1 connection between G_0 and G_1 is defined as an edge set $E = \{(v, \phi(v)) \mid v \in V_0, \phi(v) \in V_1, \text{ and } \phi : V_0 \rightarrow V_1 \text{ is a bijection}\}$. We use $G_0 \oplus G_1$ to denote $G = (V_0 \cup V_1, E_0 \cup E_1 \cup E)$. The operation “ $\oplus$ ” may generate different graphs depending on the bijection ϕ . There are some studies on the operation “ $\oplus$ ” [6, 7]. Let $G = G_0 \oplus G_1$ and x be any node in G . We use $\bar{x}$ to denote the unique node matched under ϕ .

Now, we can define the set of n -dimensional H' -graph, H'_n , as follows:

- (1) $H'_1 = \{K_2\}$, where K_2 is the complete graph with two nodes.
- (2) Assume that $G_0, G_1 \in H'_n$. Then $G = G_0 \oplus G_1$ is a graph in H'_{n+1} .

Note that some n -dimensional H' -graphs are bipartite. We can define the set of bipartite n -dimensional H' -graph, B'_n , as follows:

- (1) $B'_1 = \{K_2\}$, where K_2 is the complete graph defined on $\{a, b\}$ with bipartition $V_0 = \{a\}$ and $V_1 = \{b\}$.
- (2) For $i = 0, 1$, let G_i be a graph in B'_n with bipartition V_0^i and V_1^i . Let ϕ be a bijection between $V_0^0 \cup V_1^0$ and $V_0^1 \cup V_1^1$ such that $\phi(v) \in V_{1-i}^1$ if $v \in V_i^0$. Then $G = G_0 \oplus G_1$ is a graph in B'_{n+1} .

Every graph in H'_n is an n -regular graph with 2^n nodes, and every graph in B'_n contains 2^{n-1} nodes in each bipartition. We use N'_n to denote the set of non-bipartite graphs in H'_n . Clearly, we have $Q_n \in B'_n$.

Let G be a graph in H'_{n+1} . Then $G = G_0 \oplus G_1$ with both G_0 and G_1 in H'_n . Let u be a node in $V(G)$. Then u is a node in $V(G_i)$ for some $i = 0, 1$. We use $\bar{u}$ to denote the node in $V(G_{1-i})$ matched under ϕ . So $u = \bar{v}$ if $\bar{u} = v$.

In the following section, we give some basic properties about H'_n -graphs. In Section 3, we prove that every graph in B'_n is super spanning laceable. In Section 4, we show that every graph in N'_n is w^* -connected for every w , $1 \leq w \leq 3$, for $n \geq 3$. We also construct an N'_n -graph H and

show that H can not be 4^* -connected. In the final section, we give our concluding remark.

2 Preliminaries

Lemma 1 Assume that G is graph in N'_n . Then $n \geq 3$.

Theorem 1 [17] Let $n \geq 3$. Every graph in N'_n is hamiltonian connected and hamiltonian.

Theorem 2 [17] Every graph in B'_n is hamiltonian laceable and every graph in B'_n is hamiltonian if $n \geq 2$.

Theorem 3 [17] Let $n \geq 2$. Suppose that G is a graph in B'_n with bipartition V_0 and V_1 . Suppose that u_1 and u_2 are two distinct nodes in V_i and that v_1 and v_2 are two distinct nodes in V_{1-i} with $i \in \{0, 1\}$. Then there are two disjoint paths P_1 and P_2 of G such that (1) P_1 joins u_1 to v_1 , (2) P_2 joins u_2 to v_2 , and (3) $P_1 \cup P_2$ spans G .

Theorem 4 Let G be a graph in B'_n with bipartition V_0 and V_1 for $n \geq 2$. Suppose that z is a node in V_i and that u and v are two distinct nodes in V_{1-i} with $i \in \{0, 1\}$. Then there is a hamiltonian path of $G - \{z\}$ joining u to v .

Proof. We prove this statement by induction on n . Since Q_2 is the only graph in B'_2 , it is easy to check that this statement holds for $n = 2$. Thus, we assume that $G = G_0 \oplus G_1$ in B'_n with $n \geq 3$. We have $G_i \in B'_{n-1}$ for $i = 0, 1$. Let V_0^i and V_1^i be the bipartition of G_i for $i = 0, 1$. Without loss of generality, we assume that $V_0^0 \cup V_0^1$ and $V_1^0 \cup V_1^1$ form the bipartition of G . Let z be any node in $V_1^0 \cup V_1^1$, and let u and v be any two distinct nodes in $V_0^0 \cup V_0^1$. We need to show that there is a hamiltonian path of $G - \{z\}$ joining u to v . Without loss of generality, we assume that $z \in V_1^0$. We have the following cases:

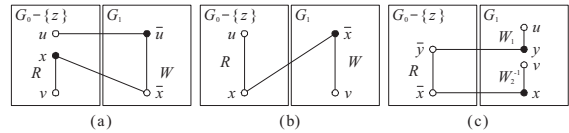


Figure 1: Illustration for Theorem 4

Case 1: $u \in V_0^0$ and $v \in V_0^0$. By induction, there is a hamiltonian path Q in $G_0 - \{z\}$ joining u to v . Without loss of generality, we write

Q as $\langle u, x, R, v \rangle$. Since $u \in V_0^0$, $x \in V_1^0$. By Theorem 2, there is a hamiltonian path W of G_1 joining the node $\bar{u} \in V_1^1$ to the node $\bar{x} \in V_0^1$. Then $\langle u, \bar{u}, W, \bar{x}, x, R, v \rangle$ is the hamiltonian path of $G - \{z\}$ joining u to v . See Figure 1(a) for an illustration.

Case 2: $u \in V_0^0$ and $v \in V_1^1$. Since $n \geq 3$, $|V_0^0| = 2^{n-1} \geq 2$. We can choose a node x in $V_0^0 - \{u\}$. By induction, there is a hamiltonian path Q in $G_0 - \{z\}$ joining u to x . Since $x \in V_0^0$, $\bar{x} \in V_1^1$. By Theorem 2, there is a hamiltonian path W of G_1 joining $\bar{x}$ to v . Then $\langle u, Q, x, \bar{x}, W, v \rangle$ is the hamiltonian path of $G - \{z\}$ joining u to v . See Figure 1(b) for an illustration.

Case 3: $u \in V_0^1$ and $v \in V_0^1$. We can choose a node x in V_1^1 . By Theorem 2, there is a hamiltonian path W in G_1 joining u to x . Without loss of generality, we write W as $\langle u, W_1, y, v, W_2, x \rangle$. Since $v \in V_0^1$, $y \in V_1^1$. By induction, there is a hamiltonian path Q in $G_0 - \{z\}$ joining the node $\bar{y} \in V_0^0$ to the node $\bar{x} \in V_0^0$. Then $\langle u, W_1, y, \bar{y}, Q, \bar{x}, x, W_2, v \rangle$ is the hamiltonian path of $G - \{z\}$ joining u to v . See Figure 1(c) for an illustration. $\square$

3 Every B'_n -graph is super spanning laceable

Let n be any positive integer. To prove that every graph in B'_n is w^* -laceable for every w , $1 \leq w \leq n$, we need the concept of spanning fan. We note that there is another Menger-type Theorem. Let u be a node of G and $S = \{v_1, v_2, \dots, v_k\}$ be a subset of $V(G)$ not including u . An (u, S) -fan is a set of disjoint paths $\{P_1, P_2, \dots, P_k\}$ of G such that P_i joins u and v_i [9]. It is proved that a graph G is k -connected if and only if there exists an (u, S) -fan between any node u and any k -subset of $V(G)$ such that $u \notin S$. With this observation, we define a *spanning fan* is a fan that spans G . Naturally, we can study $\kappa_{fan}^*(G)$ as the largest integer k such that there exists a spanning (u, S) -fan between any node u and any k -node subset S with $u \notin S$. However, we defer such a study for the following reasons.

First, let S be a cut set of a graph G . Let u be any node of $V(G) - S$. It is easy to see that there is no spanning (u, S) -fan in G . Thus, $\kappa_{fan}^*(G) < \kappa(G)$ if G is not a complete graph.

Second, let G be a bipartite graph with bipartition V_0 and V_1 and $|V_0| = |V_1|$. Let u be a vertex in V_i , $S = \{v_1, v_2, \dots, v_k\}$ be a subset of G not containing u , and $k \leq \kappa(G)$. Suppose

that $|S \cap V_{1-i}| = r$. Without loss of generality, we assume that $\{v_1, v_2, \dots, v_r\} \subset V_{1-i}$. Let $\{P_1, P_2, \dots, P_k\}$ be any spanning (u, S) -fan of G . Then $l(P_i)$ is odd if $i \leq r$, and $l(P_i)$ is even if $r < i \leq k$. Let $l(P_i) = 2t_i + 1$ if $i \leq r$ and $l(P_i) = 2t_i$ if $i > r$. For $i \leq r$, there are $t_i - 1$ nodes of P_i in V_i other than u and there are t_i nodes of P_i in V_{1-i} . For $i > r$, there are r_i nodes of P_i in V_i other than u and there are t_i nodes of P_i in V_{1-i} . Thus, $|V_i| = 1 - r + \sum_{i=1}^k t_i$ and $|V_{1-i}| = \sum_{i=1}^k t_i$. Since $|V_i| = |V_{1-i}|$, $r = 1$. Thus, $r = 1$ is a fact requirement as we study the spanning fan of bipartite graphs with equal size of bipartition.

Theorem 5 *Let n and k be any two positive integer with $k \leq n$. Let G be a graph in B'_n with bipartition V_0 and V_1 . There exists a spanning (u, S) -fan in G for any node u in V_i and any node subset S with $|S| \leq n$ such that $|S \cap V_{1-i}| = 1$ with $i \in \{0, 1\}$.*

Proof. We prove this statement by induction on n . Let $G = G_0 \oplus G_1$ in B'_n such that V_0^i and V_1^i be the bipartition of G_i for every $i = 0, 1$. Without loss of generality, we assume that $V_0^0 \cup V_0^1$ and $V_1^0 \cup V_1^1$ form the bipartition of G . Let u be any node in $V_0^0 \cup V_1^0$ and $S = \{v_1, v_2, \dots, v_k\}$ be any node subset in $G - \{u\}$ with v_1 being the unique node in $(V_1^0 \cup V_1^1) \cap S$. Without loss of generality, we assume that $u \in V_0^0$. By Theorem 2, this statement holds for $k = 1$. Thus, we assume that $k = 2$ and $n \geq 2$. By Theorem 2, there is a hamiltonian path P of G joining v_1 to v_2 . Without loss of generality, we write P as $\langle v_1, P_1, u, P_2, v_2 \rangle$. Then $\{P_1, P_2\}$ forms the spanning (u, S) -fan of G . Thus, this statement holds for $k = 2$. Moreover, this statement holds for $n = 2$. We assume that $3 \leq k \leq n$. Suppose that this statement holds for B'_{n-1} , and $G_i \in B'_{n-1}$ for $i = 0$ and 1 . Without loss of generality, we assume that $u \in G_0$. Let $T = S - \{v_1\}$. We have the following cases:

Case 1: $|T \cap V_0^0| = |T|$. Then $v_i \in V_0^0$ for every i , $2 \leq i \leq k$.

Case 1.1: $v_1 \in V_0^1$. Let $H = S - \{v_k\}$. Obviously, $H \subset G_0$, $|H \cap V_1^0| = 1$, and $|H| = k - 1$. By induction, there is a spanning (u, H) -fan $\{P_1, P_2, \dots, P_{k-1}\}$ of G_0 . Without loss of generality, we assume that P_i is joining u to v_i for every i , $1 \leq i \leq k - 1$.

Suppose that $v_k \in V(P_1)$. Without loss of generality, we write P_1 as $\langle u, Q_1, v_k, x, Q_2, v_1 \rangle$. Since $v_k \in V_0^0$, $x \in V_1^0$. (Note that $x = v_1$ if $l(Q_2) = 0$.) By Theorem 2, there is a hamiltonian path R of G_1 joining node $\bar{u} \in V_1^1$ to node $\bar{x} \in V_0^1$. We set $W_1 = \langle u, \bar{u}, R, \bar{x}, x, Q_2, v_1 \rangle$, $W_i = P_i$ for every

i , $2 \leq i \leq k-1$, and $W_k = \langle u, Q_1, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Suppose that $v_k \in V(P_i)$ for some $2 \leq i \leq k-1$. Without loss of generality, we assume that $v_k \in V(P_{k-1})$ and we write P_{k-1} as $\langle u, Q_1, v_k, x, Q_2, v_{k-1} \rangle$. Since $v_k \in V_0^0$, $x \in V_1^0$. By Theorem 2, there is a hamiltonian path R of G_1 joining node $\bar{u} \in V_1^1$ to node $\bar{x} \in V_0^1$. We set $W_i = P_i$ for every $i \in \langle k-2 \rangle$, $W_{k-1} = \langle u, \bar{u}, R, \bar{x}, x, Q_2, v_{k-1} \rangle$, and $W_k = \langle u, Q_1, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 1.2: $v_1 \in V_1^1$. We choose a node x in V_1^0 . Let $H = (T \cup \{x\}) - \{v_k\}$. So $H \subset G_0$, $|H \cap V_1^0| = 1$, and $|H| = k-1$. By induction, there is a spanning (u, H) -fan $\{P_1, P_2, \dots, P_{k-1}\}$ of G_0 . Without loss of generality, we assume that P_1 is joining u to x and P_i is joining u to v_i for every $2 \leq i \leq k-1$. We have $\bar{u} \in V_1^1$ and $\bar{x} \in V_1^0$.

Case 1.2.1: $v_k \in V(P_1)$. Without loss of generality, we write P_1 as $\langle u, Q_1, y, v_k, Q_2, x \rangle$. Since $v_k \in V_0^0$, $y \in V_1^0$ and $\bar{y} \in V_0^1$.

Suppose that $v_1 \neq \bar{u}$. By Theorem 3, there are two disjoint paths R_1 and R_2 in G_1 such that (1) R_1 joins $\bar{y}$ to v_1 , (2) R_2 joins $\bar{u}$ to $\bar{x}$, and (3) $R_1 \cup R_2$ spans G_1 . We set $W_1 = \langle u, Q_1, y, \bar{y}, R_1, v_1 \rangle$, $W_i = P_i$ for every $2 \leq i \leq k-1$, and $W_k = \langle u, \bar{u}, R_2, \bar{x}, x, Q_2^{-1}, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Suppose that $v_1 = \bar{u}$. By Theorem 4, there is a hamiltonian path R of $G_1 - \{v_1\}$ joining $\bar{y}$ to $\bar{x}$. We set $W_1 = \langle u, \bar{u} = v_1 \rangle$, $W_i = P_i$ for every $2 \leq i \leq k-1$, and $W_k = \langle u, Q_1, y, \bar{y}, R, \bar{x}, x, Q_2^{-1}, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 1.2.2: $v_k \in V(P_i)$ for some $2 \leq i \leq k-1$. Without loss of generality, we assume that $v_k \in V(P_{k-1})$ and we write P_{k-1} as $\langle u, Q_1, v_k, y, Q_2, v_{k-1} \rangle$. Since $v_k \in V_0^0$, $y \in V_1^0$ and $\bar{y} \in V_1^0$.

Suppose that $v_1 \neq \bar{u}$. By Theorem 3, there are two disjoint paths R_1 and R_2 in G_1 such that (1) R_1 joins $\bar{x}$ to v_1 , (2) R_2 joins $\bar{u}$ to $\bar{y}$, and (3) $R_1 \cup R_2$ spans G_1 . We set $W_1 = \langle u, P_1, x, \bar{x}, R_1, v_1 \rangle$, $W_i = P_i$ for every $2 \leq i \leq k-2$, $W_{k-1} = \langle u, \bar{u}, R_2, \bar{y}, y, Q_2, v_{k-1} \rangle$, and $W_k = \langle u, Q_1, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Suppose that $v_1 = \bar{u}$. By Theorem 4, there is a hamiltonian path R of $G_1 - \{v_1\}$ joining $\bar{x}$ to $\bar{y}$. We set $W_1 = \langle u, \bar{u} = v_1 \rangle$, $W_i = P_i$ for every $2 \leq i \leq k-2$, $W_{k-1} = \langle u, P_1, x, \bar{x}, R, \bar{y}, y, Q_2, v_{k-1} \rangle$, and $W_k = \langle u, Q_1, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 2: $|T \cap V_0^1| = 1$. Without loss of generality, we assume that $v_k \in V_0^1$. We have $\bar{u} \in V_1^1$.

Case 2.1: $v_1 \in V_1^0$. Let $H = S - \{v_k\}$. Obviously, $H \subset G_0$, $|H \cap V_1^0| = 1$, and $|H| = k-1$. By induction, there is a spanning (u, H) -fan $\{P_1, P_2, \dots, P_{k-1}\}$ of G_0 . Without loss of generality, we assume that P_i is joining u to v_i for every $1 \leq i \leq k-1$. By Theorem 2, there is a hamiltonian path R of G_1 joining $\bar{u}$ to v_k . We set $P_k = \langle u, \bar{u}, R, v_k \rangle$. Then $\{P_1, P_2, \dots, P_k\}$ is the spanning (u, S) -fan of G .

Case 2.2: $v_1 \in V_1^1$. By Theorem 2, there is a hamiltonian path R of G_1 joining v_1 to v_k . Without loss of generality, we write R as $\langle v_1, R_1, \bar{u}, x, R_2, v_k \rangle$. (Note that $v_1 = \bar{u}$ if $l(R_1) = 0$ and $x = v_k$ if $l(R_2) = 0$.) Since $\bar{u} \in V_1^1$, $x \in V_0^1$ and $\bar{x} \in V_1^1$. Let $H = (T \cup \{\bar{x}\}) - \{v_k\}$. Obviously, $H \subset G_0$, $|H \cap V_1^0| = 1$, and $|H| = k-1$. By induction, there is a spanning (u, H) -fan $\{P_1, P_2, \dots, P_{k-1}\}$ of G_0 . Without loss of generality, we assume that P_1 is joining u to $\bar{x}$ and P_i is joining u to v_i for every $2 \leq i \leq k-1$. We set $W_1 = \langle u, \bar{u}, R_1^{-1}, v_1 \rangle$, $W_i = P_i$ for every $2 \leq i \leq k-1$, and $W_k = \langle u, P_1, \bar{x}, x, R_2, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the (u, S) -fan of G .

Case 3: $|T \cap V_0^1| = 2$. Without loss of generality, we assume that $\{v_{k-1}, v_k\} \subset V_0^1$. We have $|V_0^0| \geq n \geq k$. We can choose a node x in $V_0^0 - \{u, v_2, v_3, \dots, v_{k-2}\}$. Obviously, $\{\bar{x}, \bar{u}\} \subset V_1^1$ with $\bar{x} \neq \bar{u}$. By Theorem 3, there are two disjoint paths R_1 and R_2 in G_1 such that (1) R_1 joins $\bar{x}$ to v_{k-1} , (2) R_2 joins $\bar{u}$ to v_k , and (3) $R_1 \cup R_2$ spans G_1 .

Case 3.1: $v_1 \in V_1^0$. Let $H = (S \cup \{x\}) - \{v_{k-1}, v_k\}$. Obviously, $H \subset G_0$, $|H \cap V_1^0| = 1$, and $|H| = k-1$. By induction, there is a spanning (u, H) -fan $\{P_1, P_2, \dots, P_{k-1}\}$ of G_0 . Without loss of generality, we assume that P_i is joining u to v_i for every $1 \leq i \leq k-1$ and P_{k-1} is joining u to x . We set $W_i = P_i$ for every $1 \leq i \leq k-2$, $W_{k-1} = \langle u, P_{k-1}, x, \bar{x}, R_1, v_{k-1} \rangle$, and $W_k = \langle u, \bar{u}, R_2, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 3.2: $v_1 \in V_1^1$ and $v_1 \in V(R_1)$. Without loss of generality, we write R_1 as $\langle \bar{x}, Q_1, v_1, y, Q_2, v_{k-1} \rangle$. Since $v_1 \in V_1^1$, $y \in V_0^1$ and $\bar{y} \in V_1^0$. Let $H = (T \cup \{x, \bar{y}\}) - \{v_{k-1}, v_k\}$. Obviously, $H \subset G_0$, $|H \cap V_1^0| = 1$, and $|H| = k-1$. By induction, there is a spanning (u, H) -fan $\{P_1, P_2, \dots, P_{k-1}\}$ of G_0 . Without loss of generality, we assume that P_1 is joining u to x , P_i is joining u to v_i for every $i \in \langle k-2 \rangle$, and P_{k-1} is joining u to $\bar{y}$. We set $W_1 = \langle u, P_1, x, \bar{x}, Q_1, v_1 \rangle$, $W_i = P_i$ for every $2 \leq i \leq k-1$, $W_{k-1} = \langle u, P_{k-1}, \bar{y}, y, Q_2, v_{k-1} \rangle$,

and $W_k = \langle u, \bar{u}, R_2, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 3.3: $v_1 \in V_1^1$ and $v_1 \in V(R_2)$. Without loss of generality, we write R_2 as $\langle \bar{u}, Q_1, v_1, y, Q_2, v_k \rangle$. Since $v_1 \in V_1^1$, $y \in V_0^1$ and $\bar{y} \in V_1^0$. Let $H = (T \cup \{x, \bar{y}\}) - \{v_{k-1}, v_k\}$. Obviously, $H \subset G_0$, $|H \cap V_1^0| = 1$, and $|H| = k - 1$. By induction, there is a spanning (u, H) -fan $\{P_1, P_2, \dots, P_{k-1}\}$ of G_0 . Without loss of generality, we assume that P_1 is joining u to x , P_i is joining u to v_i for every $2 \leq i \leq k - 2$, and P_{k-1} is joining u to $\bar{y}$. We set $W_1 = \langle u, \bar{u}, Q_1, v_1 \rangle$, $W_i = P_i$ for every $2 \leq i \leq k - 2$, $W_{k-1} = \langle u, P_1, x, \bar{x}, R_1, v_{k-1} \rangle$, and $W_k = \langle u, P_{k-1}, \bar{y}, y, Q_2, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 4: $|T \cap V_1^0| \geq 3$ and $|T \cap V_0^0| \geq 1$. We have $n \geq k = |S| \geq 5$. Without loss of generality, we assume that $A = T \cap V_0^0 = \{v_2, v_3, \dots, v_t\}$ and $B = T \cap V_1^0 = \{v_{t+1}, v_{t+2}, \dots, v_k\}$ for some $2 \leq t \leq k - 3$. Since $t \leq k - 3$ and $k \leq n$, $|A| = t - 1 \leq n - 4$ and $|B| \leq n - 2$. Since $n \geq 5$, $(n - 1)|A| + |B| \leq (n - 1)(n - 4) + (n - 2) < 2^{n-2} = |V_1^1|$. Thus, we can choose a node x in $V_1^0 - B$ such that $\bar{v}_i \notin N_{G_1}(x)$ for every $2 \leq i \leq t$. Since $2 \leq t \leq k - 3$ and $k \leq n$, $k - t + 1 \leq n - 1$. Let $H = B \cup \{\bar{u}\}$. Obviously, $H \subset G_1$, $|H \cap V_1^1| = 1$, and $|H| = k - t + 1$. By induction, there is a spanning (x, H) -fan $\{P_1, P_2, \dots, P_{k-t+1}\}$ of G_1 . Without loss of generality, we assume that P_1 is joining x to $\bar{u}$ and P_i is joining x to v_{t+i-1} for every $2 \leq i \leq k - t + 1$. Moreover, we write $P_1 = \langle x, x_1, R_1, \bar{u} \rangle$ and $P_i = \langle x, x_i, R_i, v_{t+i-1} \rangle$ for every $2 \leq i \leq k - t + 1$. Since $x \in V_1^0$, $x_i \in V_1^1$ and $\bar{x}_i \in V_0^0$ for every $1 \leq i \leq k - t + 1$. We set $C = \{\bar{x}_2, \bar{x}_3, \dots, \bar{x}_{k-t}\}$.

Case 4.1: $v_1 \in V_1^0$. Let $H' = A \cup C \cup \{v_1\}$. Obviously, $H' \subset G_0$, $|H' \cap V_1^0| = 1$, and $|H'| = k - 1$. By induction, there is a spanning (u, H') -fan $\{Q_1, Q_2, \dots, Q_{k-1}\}$ of G_0 . Without loss of generality, we assume that Q_i is joining u to v_i for every $1 \leq i \leq t$ and Q_j is joining u to $\bar{x}_{j-t+2}$ for every $t + 1 \leq j \leq k - 1$. We set $W_i = \langle u, Q_i, v_i \rangle$ for every $1 \leq i \leq t$, $W_j = \langle u, Q_j, \bar{x}_{j-t+2}, x_{j-t+2}, R_{j-t+2}, v_j \rangle$ for every $t + 1 \leq j \leq k - 1$, and $W_k = \langle u, \bar{u}, P_1^{-1}, x, P_{k-t+1}, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 4.2: $v_1 \in V_1^1$ and $v_1 \in V(P_1)$. Without loss of generality, we write P_1 as $\langle x, Z_1, y, v_1, Z_2, \bar{u} \rangle$. Since $v_1 \in V_1^1$, $y \in V_0^1$ and $\bar{y} \in V_1^0$. Let $H' = A \cup C \cap \{\bar{y}\}$. Obviously, $H' \subset G_0$, $|H' \cap V_1^0| = 1$, and $|H'| = k - 1$. By induction, there is a spanning (u, H') -fan $\{Q_1, Q_2, \dots, Q_{k-1}\}$ of G_0 . Without loss of generality, we assume

that Q_1 is joining u to $\bar{y}$, Q_i is joining u to v_i for every $2 \leq i \leq t$, and Q_j is joining u to $\bar{x}_{j-t+2}$ for every $t + 1 \leq j \leq k - 1$. We set $W_1 = \langle u, \bar{u}, Z_2^{-1}, v_1 \rangle$, $W_i = \langle u, Q_i, v_i \rangle$ for every $2 \leq i \leq t$, $W_j = \langle u, Q_j, \bar{x}_{j-t+2}, x_{j-t+2}, R_{j-t+2}, v_j \rangle$ for every $t + 1 \leq j \leq k - 1$, and $W_k = \langle u, Q_1, \bar{y}, y, Z_1^{-1}, x, P_{k-t+1}, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 4.3: $v_1 \in V_1^1$ and $v_1 \in V(P_i)$ for some $2 \leq i \leq k - t + 1$. Without loss of generality, we assume that $v_1 \in V(P_{k-t+1})$ and we write P_{k-t+1} as $\langle x, Z_1, v_1, y, Z_2, v_k \rangle$. Since $v_1 \in V_1^1$, $y \in V_0^1$ and $\bar{y} \in V_1^0$. Let $H' = A \cup C \cup \{\bar{y}\}$. Obviously, $H' \subset G_0$, $|H' \cap V_1^0| = 1$, and $|H'| = k - 1$. By induction, there is a spanning (u, H') -fan $\{Q_1, Q_2, \dots, Q_{k-1}\}$ of G_0 . Without loss of generality, we assume that Q_1 is joining u to $\bar{y}$, Q_i is joining u to v_i for every $2 \leq i \leq t$, and Q_j is joining u to $\bar{x}_{j-t+2}$ for every $t + 1 \leq j \leq k - 1$. We set $W_1 = \langle u, \bar{u}, P_1^{-1}, x, Z_1, v_1 \rangle$, $W_i = \langle u, Q_i, v_i \rangle$ for every $2 \leq i \leq t$, $W_j = \langle u, Q_j, \bar{x}_{j-t+2}, x_{j-t+2}, R_{j-t+2}, v_j \rangle$ for every $t + 1 \leq j \leq k - 1$, and $W_k = \langle u, Q_1, \bar{y}, y, Z_2, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ forms the spanning (u, S) -fan of G .

Case 5: $|T \cap V_1^0| = |T| \geq 3$. Let $H = (T \cup \{\bar{u}\}) - \{v_k\}$. Obviously, $H \subset G_1$, $|H \cap V_1^1| = 1$, and $|H| = k - 1$. By induction, there is a spanning (v_k, H) -fan $\{P_1, P_2, \dots, P_{k-1}\}$ of G_1 . Without loss of generality, we assume that P_1 is joining v_k to $\bar{u}$ and P_i is joining v_k to v_i for every $2 \leq i \leq k - 1$. Without loss of generality, we write $P_1 = \langle v_k, x_1, R_1, \bar{u} \rangle$ and write $P_i = \langle v_k, x_i, R_i, v_i \rangle$ for every $2 \leq i \leq k - 1$. Since $v_k \in V_1^0$, $x_i \in V_1^1$ and $\bar{x}_i \in V_0^0$ for every $1 \leq i \leq k - 1$. We set $C = \{\bar{x}_2, \bar{x}_3, \dots, \bar{x}_{k-t}\}$.

Case 5.1: $v_1 \in V_1^0$. Let $H' = C \cup \{v_1\}$. Obviously, $H' \subset G_0$, $|H' \cap V_1^0| = 1$, and $|H'| = k - 1$. By induction, there is a spanning (u, H') -fan $\{Q_1, Q_2, \dots, Q_{k-1}\}$ of G_0 . Without loss of generality, we assume that Q_1 is joining u to v_1 and Q_i is joining u to $\bar{x}_i$ for every $2 \leq i \leq k - 1$. We set $W_1 = Q_1$, $W_i = \langle u, Q_i, \bar{x}_i, x_i, R_i, v_i \rangle$ for every $2 \leq i \leq k - 1$, and $W_k = \langle u, \bar{u}, P_1^{-1}, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 5.2: $v_1 \in V_1^1$ and $v_1 \in V(P_1)$. Without loss of generality, we write $P_1 = \langle v_k, Z_1, y, v_1, Z_2, \bar{u} \rangle$. Since $v_1 \in V_1^1$, $y \in V_0^1$ and $\bar{y} \in V_1^0$. Let $H' = C \cup \{\bar{y}\}$. Obviously, $H' \subset G_0$, $|H' \cap V_1^0| = 1$, and $|H'| = k - 1$. By induction, there is a spanning (u, H') -fan $\{Q_1, Q_2, \dots, Q_{k-1}\}$ of G_0 . Without loss of generality, we assume that Q_1 is joining u to $\bar{y}$ and Q_i is joining u to $\bar{x}_i$ for every $2 \leq i \leq k - 1$. We set $W_1 = \langle u, \bar{u}, Z_2^{-1}, v_1 \rangle$,

$W_i = \langle u, Q_i, \bar{x}_i, x_i, R_i, v_i \rangle$ for every $2 \leq i \leq k-1$, and $W_k = \langle u, Q_1, \bar{y}, y, Z_1^{-1}, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G .

Case 5.3: $v_1 \in V_1^1$ and $v_1 \in V(P_i)$ for some $2 \leq i \leq k-1$. Without loss of generality, we assume that $v_1 \in V(P_{k-1})$ and write $P_{k-1} = \langle v_k, x_{k-1}, Z_1, v_1, y, Z_2, v_{k-1} \rangle$. Since $v_1 \in V_1^1$, $y \in V_0^1$ and $\bar{y} \in V_1^0$. Let $H' = C \cup \{\bar{y}\}$. Obviously, $H' \subset G_0$, $|H' \cap V_1^0| = 1$, and $|H'| = k-1$. By induction, there is a (u, H') -fan $\{Q_1, Q_2, \dots, Q_{k-1}\}$ of G_0 . Without loss of generality, we assume that Q_1 is joining u to $\bar{y}$ and Q_i is joining u to $\bar{x}_i$ for every $2 \leq i \leq k-1$. We set $W_1 = \langle u, Q_{k-1}, \bar{x}_{k-1}, x_{k-1}, Z_1, v_1 \rangle$, $W_i = \langle u, Q_i, \bar{x}_i, x_i, R_i, v_i \rangle$ for every $2 \leq i \leq k-2$, $W_{k-1} = \langle u, Q_1, \bar{y}, y, Z_2, v_{k-1} \rangle$, and $W_k = \langle u, \bar{u}, P_1^{-1}, v_k \rangle$. Then $\{W_1, W_2, \dots, W_k\}$ is the spanning (u, S) -fan of G . $\square$

Theorem 6 Every graph in B'_n is super spanning laceable for $n \geq 1$.

Proof. Suppose that $G = G_0 \oplus G_1$ in B'_n with bipartition V_0 and V_1 . Let u be any node in V_0 and v be any node in V_1 . We need to show there is a k^* -container of G between u and v for every positive integer k with $k \leq n$. By Theorem 2, there is a 1^* -container of G joining u to v . Thus, we assume that $k \geq 2$ and $n \geq 2$. Since $k \leq n$ and $|N_G(v)| = n$, we can choose $(k-1)$ distinct nodes $x_1, x_2, \dots, x_{k-1}$ in $N_G(v) - \{u\}$. Since v is in V_1 , x_i is in $V_0 - \{u\}$ for $i = 1$ to $k-1$. We set $S = \{v, x_1, x_2, \dots, x_{k-1}\}$. By Theorem 5, there is a spanning (u, S) -fan $\{R_1, R_2, \dots, R_k\}$ of G . Without loss of generality, we assume that R_1 is joining u to v and R_i is joining u to x_{i-1} for every $2 \leq i \leq k$. We set $P_1 = R_1$ and $P_i = \langle u, R_i, x_{i-1}, v \rangle$ for every $2 \leq i \leq k$. Then $\{P_1, P_2, \dots, P_k\}$ is the k^* -container of G between u and v . $\square$

4 On the w^* -connectedness of N'_n -graphs

4.1 Every graph in N'_n is 3^* -connected

Lemma 2 According to isomorphism, there is only one graph in N'_3 . Moreover, this graph is 3^* -connected.

Proof. By brute force, we can check the graph T in Figure 2 is the only graph in N'_3 .

Let x and y be two distinct nodes of T . By the symmetric of T , we can assume that $x = 0$ and

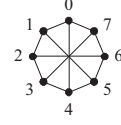


Figure 2: The only graph T in N'_3

$y \in \{1, 2, 3, 4\}$. The 3^* -containers $\{P_1, P_2, P_3\}$ of T between x and y are list below:

$y = 1$	$\{P_1 = \langle 0, 1 \rangle, P_2 = \langle 0, 4, 3, 2, 1 \rangle, P_3 = \langle 0, 7, 6, 5, 1 \rangle\}$
$y = 2$	$\{P_1 = \langle 0, 1, 2 \rangle, P_2 = \langle 0, 7, 3, 2 \rangle, P_3 = \langle 0, 4, 5, 6, 2 \rangle\}$
$y = 3$	$\{P_1 = \langle 0, 4, 3 \rangle, P_2 = \langle 0, 7, 3 \rangle, P_3 = \langle 0, 1, 5, 6, 2, 3 \rangle\}$
$y = 4$	$\{P_1 = \langle 0, 4 \rangle, P_2 = \langle 0, 1, 2, 3, 4 \rangle, P_3 = \langle 0, 7, 6, 5, 4 \rangle\}$

Thus, T is 3^* -connected. $\square$

Let $n \geq 3$. Let $G = G_0 \oplus G_1 \in N'_{n+1}$ with $G_0 \in H'_n$ and $G_1 \in H'_n$. Depending on G_0 and G_1 is bipartite or not, we prove that $G = G_0 \oplus G_1$ is 3^* -connected with the following lemmas.

Lemma 3 Let $n \geq 3$. Assume that $G = G_0 \oplus G_1$ in N'_{n+1} with both G_0 and G_1 in N'_n . Then G is 3^* -connected.

Proof. Let u and v be any two distinct nodes of G . We need to construct a 3^* -container of G between u and v .

Case 1: $u, v \in G_0$. By Theorem 1, there is a 2^* -container $\{P_1, P_2\}$ of G_0 between u and v . By Theorem 1 again, there is a hamiltonian path P of G_1 joining $\bar{u}$ to $\bar{v}$. We set P_3 as $\langle u, \bar{u}, P, \bar{v}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 2: $u \in G_0$ and $v \in G_1$ with $\bar{u} = v$. Since there are 2^n nodes in G_0 and $2^n > 3$ for $n \geq 3$, we can choose two distinct nodes x and y in $G_0 - \{u\}$. By Theorem 1, there is a hamiltonian path R of G_0 joining x to y . Again, there is a hamiltonian path W of G_1 joining $\bar{x}$ to $\bar{y}$. We write $R = \langle x, R_1, u, R_2, y \rangle$ and $W = \langle \bar{x}, W_1, v, W_2, \bar{y} \rangle$. We set $P_1 = \langle u, R_1^{-1}, x, \bar{x}, W_1, v \rangle$, $P_2 = \langle u, R_2, y, \bar{y}, W_2^{-1}, v \rangle$, and $P_3 = \langle u, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 3: $u \in G_0$ and $v \in G_1$ with $\bar{u} \neq v$. Since there are 2^n nodes in G_0 , we choose a node x in $G_0 - \{u, \bar{v}\}$. By Theorem 1, there is a hamiltonian path R of G_0 joining x to $\bar{v}$. Again, there is a hamiltonian path W of G_1 joining $\bar{x}$ to $\bar{u}$. We write $R = \langle x, R_1, u, R_2, \bar{v} \rangle$ and $W = \langle \bar{x}, W_1, v, W_2, \bar{u} \rangle$. We set $P_1 = \langle u, \bar{u}, W_2^{-1}, v \rangle$, $P_2 = \langle u, R_1^{-1}, x, \bar{x}, W_1, v \rangle$, and $P_3 = \langle u, R_2, \bar{v}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v . $\square$

Lemma 4 Let $n \geq 3$. Assume that $G = G_0 \oplus G_1$ in N'_{n+1} with G_0 in B'_n and G_1 in N'_n . Then G is 3^* -connected.

Proof. Let V_0 and V_1 be the bipartition of G_0 . Let u and v be any two distinct nodes of G . We need to construct a 3^* -container of G between u and v .

Case 1: $u, v \in G_0$. By Theorem 2, there is a 2^* -container $\{P_1, P_2\}$ of G_0 between u and v . By Theorem 1, there is a hamiltonian path P of G_1 joining $\bar{u}$ to $\bar{v}$. We set $P_3 = \langle u, \bar{u}, P, \bar{v}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 2: $u, v \in G_1$. Without loss of generality, we assume that $\bar{u} \in V_0$.

Case 2.1: $\bar{v} \in V_0$. Since there are 2^{n-1} nodes in V_1 and $2^{n-1} \geq 4$ for $n \geq 3$, we can choose two distinct nodes x and y in V_1 . By Theorem 1, there is a hamiltonian path R of G_1 joining $\bar{x}$ to $\bar{y}$. Without loss of generality, we write $R = \langle \bar{x}, R_1, u, R_2, v, R_3, \bar{y} \rangle$. By Theorem 3, there are two disjoint paths T_1 and T_2 of G_0 such that (1) T_1 joins $\bar{u}$ to y , (2) T_2 joins x to $\bar{v}$, and (3) $T_1 \cup T_2$ spans G_1 . We set $P_1 = \langle u, R_2, v \rangle$, $P_2 = \langle u, R_1^{-1}, \bar{x}, x, T_2, \bar{v}, v \rangle$, and $P_3 = \langle u, \bar{u}, T_1, y, \bar{y}, R_3^{-1}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 2.2: $\bar{v} \in V_1$. By Theorem 1, there is a 2^* -container $\{P_1, P_2\}$ of G_1 between u and v . By Theorem 2, there is a hamiltonian path P of G_0 joining $\bar{u}$ to $\bar{v}$. We set $P_3 = \langle u, \bar{u}, P, \bar{v}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 3: $u \in G_0$ and $v \in G_1$ with $\bar{u} \neq v$. By Theorem 2, there is a hamiltonian cycle C of G_0 . Without loss of generality, we write $C = \langle u, R_1, \bar{v}, x, R_2, u \rangle$. By Theorem 1, there is a hamiltonian path T of G_1 joining $\bar{u}$ to $\bar{x}$. Without loss of generality, we write $T = \langle \bar{u}, T_1, v, T_2, \bar{x} \rangle$. We set $P_1 = \langle u, R_1, \bar{v}, v \rangle$, $P_2 = \langle u, \bar{u}, T_1, v \rangle$, and $P_3 = \langle u, R_2^{-1}, x, \bar{x}, T_2^{-1}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 4: $u \in G_0$ and $v \in G_1$ with $\bar{u} = v$. Without loss of generality, we assume that $u \in V_0$. We can choose a node x in $V_0 - \{u\}$ and a node y in V_1 . By Theorem 2, there is a hamiltonian path R of G_0 joining x to y . By Theorem 1, there is a hamiltonian path T of G_1 joining $\bar{x}$ to $\bar{y}$. Without loss of generality, we write $R = \langle x, R_1, u, R_2, y \rangle$ and $T = \langle \bar{x}, T_1, v, T_2, \bar{y} \rangle$. We set $P_1 = \langle u, v \rangle$, $P_2 = \langle u, R_1^{-1}, x, \bar{x}, T_1, v \rangle$, and $P_3 = \langle u, R_2, y, \bar{y}, T_2^{-1}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v . $\square$

Lemma 5 Assume that $G = G_0 \oplus G_1$ in N'_{n+1} with both G_0 and G_1 in B'_n for $n \geq 2$. Then G is 3^* -connected.

Proof. Let V_0^i and V_1^i be the bipartition of G_i for $i = 0, 1$. Let u and v be two distinct nodes of G . Without loss of generality, we assume that $u \in V_0^0$ and $\bar{u} \in V_1^1$. We need to construct a 3^* -container of G between u and v .

Case 1: $v \in V_0^0 \cup V_1^0$ and $\bar{v} \in V_0^1$. By Theorem 2, there is a 2^* -container $\{P_1, P_2\}$ of G_0 between u and v . By Theorem 2, there is a hamiltonian path P of G_1 joining $\bar{u}$ to $\bar{v}$. We set $P_3 = \langle u, \bar{u}, P, \bar{v}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 2: $v \in V_0^0$ and $\bar{v} \in V_1^1$. Since $u \in V_0^0$, $\bar{u} \in V_1^1$, $v \in V_0^0$, and $\bar{v} \in V_1^1$, we can choose a node x in V_1^0 such that $\bar{x} \in V_0^1$ and choose a node y in V_0^0 such that $\bar{y} \in V_0^1$. By Theorem 2, there is a hamiltonian path R of G_0 joining x to y . Without loss of generality, we write $R = \langle x, R_1, p, R_2, q, R_3, y \rangle$ where $\{p, q\} = \{u, v\}$. Without loss of generality, we assume that $p = u$ and $q = v$. By Theorem 3, there are two disjoint paths T_1 and T_2 of G_1 such that (1) T_1 joins $\bar{x}$ to $\bar{v}$, (2) T_2 joins $\bar{u}$ to $\bar{y}$, and (3) $T_1 \cup T_2$ spans G_1 . We set $P_1 = \langle u, R_2, v \rangle$, $P_2 = \langle u, R_1^{-1}, x, \bar{x}, T_1, \bar{v}, v \rangle$, $P_3 = \langle u, \bar{u}, T_2, \bar{y}, y, R_3^{-1}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 3: $v \in V_1^0$ and $\bar{v} \in V_1^1$. Since $u \in V_0^0$ and $\bar{u} \in V_1^1$, $v \in V_1^0$, and $\bar{v} \in V_1^1$, we can choose a node x in V_0^1 such that $\bar{x} \in V_0^1$ and choose a node y in V_0^0 such that $\bar{y} \in V_0^1$. By Theorem 2, there is a hamiltonian path R of G_0 joining x to y . Without loss of generality, we write $R = \langle x, R_1, p, R_2, q, R_3, y \rangle$ where $\{p, q\} = \{u, v\}$. Without loss of generality, we assume that $p = u$ and $q = v$. By Theorem 3, there are two disjoint paths T_1 and T_2 of G_1 such that (1) T_1 joins $\bar{x}$ to $\bar{v}$, (2) T_2 joins $\bar{u}$ to $\bar{y}$, and (3) $T_1 \cup T_2$ spans G_1 . We set $P_1 = \langle u, R_2, v \rangle$, $P_2 = \langle u, R_1^{-1}, x, \bar{x}, T_1, \bar{v}, v \rangle$, $P_3 = \langle u, \bar{u}, T_2, \bar{y}, y, R_3^{-1}, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 4: $v \in V_0^1 \cup V_1^1$ and $\bar{u} \neq v$.

Case 4.1: $\bar{v} \in V_0^0$. Since $u \in V_0^0$, $\bar{u} \in V_1^1$, and $\bar{v} \in V_0^0$, we can choose a node $x \in V_0^1$ such that $\bar{x} \in V_0^1$. By Theorem 2, there is a hamiltonian path R of G_0 joining x to $\bar{v}$. Again, by Theorem 2, there is a hamiltonian path T of G_1 joining $\bar{x}$ to $\bar{u}$. Write $R = \langle x, R_1, u, R_2, \bar{v} \rangle$ and $T = \langle \bar{x}, T_1, v, T_2, \bar{u} \rangle$. We set $P_1 = \langle u, \bar{u}, T_2^{-1}, v \rangle$, $P_2 = \langle u, R_2, \bar{v}, v \rangle$, and $P_3 = \langle u, R_1^{-1}, x, \bar{x}, T_1, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 4.2: $\bar{v} \in V_1^0$ and $v \in V_0^1$. Since $u \in V_0^0$, $\bar{u} \in V_1^1$, $v \in V_0^1$, and $\bar{v} \in V_1^0$, we can choose a node $x \in V_0^0$ such that $\bar{x} \in V_0^1$. By Theorem 2, there is a hamiltonian path R of G_0 joining x to $\bar{v}$, and there is a hamiltonian path T of G_1 joining $\bar{x}$ to $\bar{u}$. We write $R = \langle x, R_1, u, R_2, \bar{v} \rangle$ and $T = \langle \bar{x}, T_1, v, T_2, \bar{u} \rangle$. We set $P_1 = \langle u, \bar{u}, T_2^{-1}, v \rangle$, $P_2 = \langle u, R_2, \bar{v}, v \rangle$, and $P_3 = \langle u, R_1^{-1}, x, \bar{x}, T_1, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 4.3: $\bar{v} \in V_1^0$ and $v \in V_1^1$. Since $u \in V_0^0$, $\bar{u} \in V_1^1$, and $v \in V_1^1$, we can choose a node $x \in V_0^0$ such that $\bar{x} \in V_0^1$. By Theorem 2, there is a hamiltonian path R of G_0 joining x to $\bar{v}$, and there is a hamiltonian path T of G_1 joining $\bar{x}$ to $\bar{u}$. We write $R = \langle x, R_1, u, R_2, \bar{v} \rangle$ and $T = \langle \bar{x}, T_1, v, T_2, \bar{u} \rangle$. We set $P_1 = \langle u, \bar{u}, T_2^{-1}, v \rangle$, $P_2 = \langle u, R_2, \bar{v}, v \rangle$, and $P_3 = \langle u, R_1^{-1}, x, \bar{x}, T_1, v \rangle$. Then $\{P_1, P_2, P_3\}$ is the 3^* -container of G between u and v .

Case 5: $v = \bar{u}$. Since $u \in V_0^0$ and $\bar{u} \in V_1^1$, we can choose a node $x \in V_0^0$ such that $\bar{x} \in V_0^1$ and choose a node $y \in V_1^0$ such that $\bar{y} \in V_1^1$. By Theorem 2, there is a hamiltonian path R of G_0 joining x to y , and there is a hamiltonian path T of G_1 joining $\bar{x}$ to $\bar{y}$. Without loss of generality, we write that $R = \langle x, R_1, u, R_2, y \rangle$ and $T = \langle \bar{x}, T_1, v, T_2, \bar{y} \rangle$. We set $P_1 = \langle u, v \rangle$, $P_2 = \langle u, R_1^{-1}, x, \bar{x}, T_1, v \rangle$, and $P_3 = \langle u, R_2, y, \bar{y}, T_2^{-1}, v \rangle$. Then $\{P_1, P_2, P_3\}$ forms the 3^* -container of G between u and v . $\square$

With Lemma 2, Lemma 3, Lemma 4, and Lemma 5, we have the following theorem:

Theorem 7 *Every graph in N'_n is 3^* -connected.*

4.2 An N'_n -graph H is not 4^* -connected

We say that $u = u_n u_{n-1} \dots u_2 u_1$ is an n -bit binary string if $u_i \in \{0, 1\}$ for every $1 \leq i \leq n$. For $1 \leq i \leq n$, we use $(u)^i$ to denote the binary string, $v_n v_{n-1} \dots v_2 v_1$, such that $v_i = 1 - u_i$ and $v_j = u_j$ for every $j \neq i$. Moreover, we use $(u)_i$ to denote u_i . The *Hamming weight* of an n -bit binary strings $u = u_n u_{n-1} \dots u_2 u_1$, $w(u)$, is $\sum_{i=1}^n u_i$. The n -dimensional hypercube, Q_n , consists of all n -bit binary strings as its nodes. Two nodes $u = u_n u_{n-1} \dots u_2 u_1$ and $v = v_n v_{n-1} \dots v_2 v_1$ of Q_n are adjacent if and only if $v = (u)^i$ for some $i \in \{1, 2, \dots, n\}$. Note that Q_n is a bipartite graph with bipartition $\{u \mid w(u) \text{ is even}\}$ and $\{u \mid w(u) \text{ is odd}\}$. Let Q_n^i be the subgraph of Q_n induced by $\{u \in V(Q_n) \mid (u)_n = i\}$ for $i \in \{0, 1\}$. Then Q_n^i is isomorphic to Q_{n-1} . By the definition of Q_n , $Q_n \in B'_n$. Let $n \geq 4$ and let $e = \underbrace{00 \dots 0}_n$ be a node in Q_n . We set $v = (e)^1$, $p = (e)^n$, and $q = ((e)^1)^n$.

Let H be the graph with $V(H) = V(Q_n)$ and $E(H) = (E(Q_n) - \{(e, p), (v, q)\}) \cup \{(e, q), (v, p)\}$. Obviously, $H - \{(e, q), (v, p)\}$ is a bipartite graph with bipartition $A = \{x \mid w(x) \text{ is even}\}$ and $B = \{x \mid w(x) \text{ is odd}\}$. Moreover, H is in N'_n and $H = Q_n^0 \oplus Q_n^1$ for some 1-1 connection ϕ . We will show that H is not k^* -connected for $k \geq 4$.

Suppose that there is a k^* -container $C = \{P_1, P_2, \dots, P_k\}$ of H between e and q for some $k \geq 4$. We have the following cases:

Case 1: $(e, q) \in \cup_{i=1}^k P_i$ and $(v, p) \in \cup_{i=1}^k P_i$. Without loss of generality, we assume that $(e, q) \in P_1$. Thus, $P_1 = \langle e, q \rangle$. Again, we can assume without loss of generality that $(v, p) \in P_2$. Obviously, the number of nodes in P_2 is $2t_2$ for some integer t_2 and the number of nodes in P_i is $2t_i + 1$ for some integer t_i for every $3 \leq i \leq k$. Therefore, there are t_2 nodes of $V(P_1) \cap B$ and $(t_2 - 2)$ nodes of $V(P_1) \cap A$ other than e and q , and there are t_i nodes of $V(P_i) \cap B$ and $(t_i - 1)$ nodes of $V(P_i) \cap A$ other than e and q for every $3 \leq i \leq k$. As a consequence, $|A| = \sum_{i=2}^k t_i + 2 - k$ and $|B| = \sum_{i=2}^k t_i$. Thus, $|A| \neq |B|$.

Case 2: $(e, q) \in \cup_{i=1}^k P_i$ and $(v, p) \notin \cup_{i=1}^k P_i$. Without loss of generality, we assume that $(e, q) \in P_1$. Obviously, the number of nodes in P_i is $(2t_i + 1)$ for some integer t_i for every $2 \leq i \leq k$. Moreover, there are t_i nodes of $V(P_i) \cap B$, and $(t_i - 1)$ nodes of $V(P_i) \cap A$ other than e and q for every $2 \leq i \leq k$. As a consequence, $|A| = \sum_{i=2}^k t_i + 3 - k$ and $|B| = \sum_{i=2}^k t_i$. Thus, $|A| \neq |B|$.

Case 3: $(e, q) \notin \cup_{i=1}^k P_i$ and $(v, p) \in \cup_{i=1}^k P_i$. Without loss of generality, we assume that $(v, p) \in P_1$. Obviously, the number of nodes in P_1 is $2t_1$ for some integer t_1 , and the number of nodes in P_i is $(2t_i + 1)$ for some integer t_i for every $2 \leq i \leq k$. Moreover, there are t_1 nodes of $V(P_1) \cap B$ and $(t_1 - 2)$ nodes of $V(P_1) \cap A$ other than e and q , and there are t_i nodes of $V(P_i) \cap B$ and $(t_i - 1)$ nodes of $V(P_i) \cap A$ other than e and q for every $2 \leq i \leq k$. As a consequence, $|A| = \sum_{i=1}^k t_i + 1 - k$ and $|B| = \sum_{i=1}^k t_i$. Thus, $|A| \neq |B|$.

Case 4: $(e, q) \notin \cup_{i=1}^k P_i$ and $(v, p) \notin \cup_{i=1}^k P_i$. Obviously, the number of nodes in P_i is $(2t_i + 1)$ for some integer t_i for every $1 \leq i \leq k$. Moreover, there are t_i nodes of $V(P_i) \cap B$, and $(t_i - 1)$ nodes of $V(P_i) \cap A$ other than e and q for every $1 \leq i \leq k$. As a consequence, $|A| = \sum_{i=1}^k t_i + 2 - k$ and $|B| = \sum_{i=1}^k t_i$. Thus, $|A| \neq |B|$.

With Case 1, Case 2, Case 3, and Case 4, C is not a k^* -container of H between e and q . Thus, H is not k^* -connected for any k , $4 \leq k \leq n$.

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